

STATISTICAL ANALYSIS OF FACTORS INFLUENCING AUTOMOBILE FUEL EFFICIENCY

Probability and Statistics



Abstract

Fuel efficiency in automobiles is a vital factor influencing environmental sustainability and economic costs, especially amid rising energy demands. This report examines fuel efficiency using the Auto-MPG dataset, originally sourced from the StatLib library at Carnegie Mellon University and featured in the American Statistical Association Exposition, covering automobiles from 1970 to 1982. The study focuses on numerical variables such as miles per gallon (mpg), vehicle weight, engine displacement, and horsepower. It outlines techniques for sampling and grouping data, methods for analyzing descriptive and numerical variables, and stages of data interpretation. The purpose is to derive meaningful conclusions on factors affecting fuel efficiency, providing insights with implications for modern automotive design and policy.

Declaration

I, F.S. Noufer, a student at SLIIT City University, am currently enrolled in the Probability and Statistics module. I declare this report on “Statistical Analysis of Factors Influencing Automobile Fuel Efficiency” to be my original work for continuous assessments in the same module. The data used in this report were obtained from the UCI Machine Learning Repository and have been properly acknowledged. I guarantee the confidentiality and anonymity of any data collected where applicable. I confirm that such data will not be used outside this study. Additionally, I take full responsibility for any effects or results from this report.
[(13 October , 2025)]

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Table of Abbreviations

MPG-Miles Per Gallon

Chapter 1 – Introduction

1.1 Background

Fuel efficiency has remained a key factor in car production and purchasing. It has propelled the evolution of the industry for centuries. The oil shocks that were felt around the world in the 1970s, especially the 1973 embargo, served to create a radical shift in what motor manufacturers valued. Manufacturers began valuing fuel efficiency more than brute force. This shift was a watershed moment in history as rising fuel costs and environmental concerns spurred the manufacture of cars in a manner that would achieve the highest mileage per gallon (mpg), a significant measure of performance and environmental sustainability. Understanding why fuel was being consumed was needed both for economic reasons as well as curtailing the environmental damage wrought by transportation, a priority which persists to this day.

The Auto-MPG dataset, provided by the UCI Machine Learning Repository and initially compiled for the 1983 American Statistical Association Exposition, provides a welcome glimpse into this period of turmoil. Spanning the 1970 to 1982 period, it contains data on 398 cars, providing a high-resolution historical record of car characteristics and their fuel efficiency. Significant variables are mpg as the primary measurement, and independent variables such as the number of cylinders, engine size (in cubic inches), horsepower, car weight (in pounds), acceleration (in seconds), model year, and origin (indicated as US, Europe, or Japan). Although outdated, this data forms a solid foundation for statistical analysis based on design development spurred by economic and legislative demands of the period.

This analysis explores how various car attributes, such as weight and engine size, relate to fuel efficiency during the 1970s and early 1980s. It investigates trends, like the shift to lighter cars and smaller engines, which likely responded to fuel shortages and can be measured with statistical correlations. The goal is to draw lessons from this past data to inform today's car design and fuel efficiency standards, even though the dataset predates modern technologies like hybrids or electric vehicles. The focus on basic principles such as the impact of weight and engine size provides valuable insights for developing greener car technology. This historical context, supported by statistical analysis, sets the foundation for the research questions explored in the following sections.

1.2 Problem Identification

Car fuel efficiency was a key concern during the 1970s and 1980s with global oil crises, prompting automobile makers to rethink design priorities. At that time, poorly efficient vehicles ballooned fuel costs and produced more environmental impact with increased emissions and resource use. Historic challenges sparked design changes that still influence current focus on sustainable transport. The Auto-MPG dataset, with actual data about the cars between 1970 and 1982, offers a useful tool for this report to analyze variables such as horsepower, weight, and engine size by statistical methods.

This study seeks to understand how such vehicle traits shaped fuel economy at the time, looking at past trends rather than the present market. Based on a 50-observation sample, it draws on data sampling, aggregation, and numerical techniques in exploring the topography of efficiency. It seeks to offer explanations on how such factors impacted past car performance as a basis to analyze past design decisions without necessarily applying them to present scenarios.

1.3 Significance of research

This research evaluates the factors influencing automobile fuel efficiency during the 1970s and early 1980s, using the Auto-MPG dataset. By conducting this analysis successfully, it is possible to identify which vehicle attributes, such as weight and engine size, most affected miles per gallon (mpg) during a period shaped by oil crises. Furthermore, carrying out this research will help uncover how design choices impacted fuel economy and how these trends evolved over time. This study determines whether past engineering efforts adequately addressed fuel efficiency or if further focus was needed, providing a historical perspective.

Success of this research marks the first step toward understanding automotive design trade-offs during this era. New statistical insights derived from this analysis can be utilized to explore what drove efficiency challenges and to assess their historical implications. The major importance of this research lies in generating new statistical information that can lead to further studies on automotive history, offering a foundation for evaluating past practices without assuming relevance to today's market.

1.4 Objectives of the study

The primary goal of this study is to identify the key factors influencing automobile fuel efficiency during the 1970s and early 1980s using the Auto-MPG dataset. The specific objectives this research aims to achieve are as follows:

- Determine the distribution of miles per gallon (mpg) across the sampled vehicles.
- Assess the variation in engine displacement among the vehicles.
- Analyze the weight distribution of vehicles in the dataset.

- Examine the impact of vehicle weight on fuel efficiency.
- Investigate the effect of engine horsepower on fuel efficiency.
- Evaluate how the number of cylinders influences fuel consumption patterns.
- Assess the overall fuel efficiency levels of vehicles from 1970-1982 using statistical methods.
- Draw conclusions from the analyzed data to address whether vehicle design effectively tackled fuel efficiency challenges and identify the primary factors influencing mpg during this period.

1.5 Chapter Framework

This chapter introduces the study titled "Statistical Analysis of Factors Influencing Automobile Fuel Efficiency – Insights from the Auto-MPG Dataset," designed to explore historical fuel efficiency trends from 1970-1982 using statistical techniques as a hands-on learning exercise.

1.1 Background – Establishes the context by discussing the critical role of fuel efficiency during the 1970s oil crises, introducing the Auto-MPG dataset as a key historical resource, and highlighting its value for statistical learning.

1.2 Problem Identification – Outlines the challenge of using statistical methods to analyze how vehicle attributes historically affected fuel efficiency, presenting it as a practical learning opportunity.

1.3 Significance of Research – Emphasizes the study's importance in enhancing statistical skills, exploring variable relationships, and providing a foundation for future historical data analyses.

1.4 Objectives of the Study – Specifies the goals, including analysis of the Auto-MPG dataset using sampling, examine relationships between vehicle features and fuel efficiency, test their significance, and interpret the results.

1.5 Chapter Framework – Offers an overview of the chapter's layout, focusing on its role in using historical data to build statistical expertise.

Chapter 2 - Literature Review

Understanding the factors influencing automobile fuel efficiency has been a subject of extensive research since the oil crises of the 1970s [von Hippel and Levi, 1983]. This chapter examines previous studies to contextualize the current analysis of the Auto-MPG dataset, highlighting historical trends, methodological approaches, and gaps that the present study aims to address.

Research by Quinlan [1993] utilized the Auto-MPG dataset for machine learning applications, specifically predicting miles per gallon (mpg) values based on vehicle attributes such as engine displacement, weight, and model year. His findings established that these variables were significant predictors of fuel economy, providing a foundational statistical framework for analyzing automotive efficiency during the 1970s and 1980s [Quinlan, 1993]. This work underscores the relevance of the dataset for quantitative analysis, aligning with the current study's focus on numerical data.

Studies from the 1970s and 1980s documented the automotive industry's response to fuel scarcity following the 1973 oil embargo [American Statistical Association, 1983]. Research presented at the 1983 American Statistical Association Exposition analyzed trends indicating that vehicle weight reduction and engine downsizing were primary strategies employed by manufacturers to improve fuel efficiency [American Statistical Association, 1983]. These studies identified strong negative correlations between vehicle weight and mpg, as well as between engine displacement and fuel consumption, findings that remain pertinent for historical context and are testable with the Auto-MPG data.

Comparative analyses of vehicles from different origins—namely the United States, Europe, and Japan—revealed distinct approaches to fuel efficiency. Japanese manufacturers during this period emphasized smaller engines and lighter chassis, achieving higher mpg values, while American manufacturers initially maintained larger vehicles but gradually reduced engine sizes in response to regulatory pressures [Sivak, 2019]. European manufacturers occupied a middle ground, balancing performance with efficiency through innovative engineering [von Hippel and Levi, 1983]. The Auto-MPG dataset's inclusion of the 'origin' variable allows for similar comparative analysis, enhancing the study's scope.

More recent research has extended these insights into modern contexts. Greene [1992] explored how fuel economy standards, such as the Corporate Average Fuel Economy (CAFE) regulations introduced in 1975, influenced long-term efficiency trends, noting a shift toward smaller engines and lighter materials. Similarly, DeCicco [2010] highlighted the enduring impact of weight and engine size on fuel efficiency, suggesting that historical data can inform contemporary policy. However, these studies often lack detailed statistical methodologies applied to specific historical datasets like Auto-MPG, presenting a gap that this research addresses [Greene, 1992; DeCicco, 2010].

Further insights from U.S. Environmental Protection Agency [2025] reports indicate that fuel efficiency improvements have been a continuous focus, with historical data serving as a benchmark for modern standards. The General Insurance [2024] analysis of fuel-efficient vehicles since 1975 reinforces the importance of engine design, while

U.S. Department of Energy [2007] trends highlight the role of weight reduction. EBSCO [n.d.] and Quora [2022] discussions provide additional perspectives on industry shifts and consumer expectations, though they lack the statistical depth of primary research.

What differentiates the current study is its application of statistical methodologies—emphasizing sampling techniques, grouping strategies, and inferential statistics—to the Auto-MPG dataset [Quinlan, 1993]. This approach aims to derive quantitative insights into how vehicle attributes influenced fuel economy between 1970 and 1982, a transformative era marked by energy crises and regulatory changes [von Hippel and Levi, 1983]. By building on previous research, this analysis provides a deeper understanding of historical efficiency trends, offering a basis for recommendations that could guide future studies into modern automotive technologies [U.S. Environmental Protection Agency, 2025].

Chapter 3 - Theory and Methodology

3.1 Research Design

This study analyzes factors influencing automobile fuel efficiency using the Auto-MPG dataset from the UCI Machine Learning Repository. The research employs quantitative statistical methods to examine relationships between vehicle attributes and miles per gallon (mpg) during the 1970-1982 period.

The sampling method selected is **simple random sampling** from the complete dataset of 398 vehicles. A sample size of 50 vehicles was chosen to balance statistical validity with computational manageability while ensuring adequate representation across model years and countries of origin.

The analysis framework combines descriptive statistics, Parametric tests analysis, hypothesis testing, and chi-squared tests to evaluate the relationships between independent variables (cylinders, displacement, horsepower, weight, acceleration, model year, origin) and the dependent variable (mpg).

3.2 Data Collection Method

The Auto-MPG dataset was obtained from the UCI Machine Learning Repository. The original dataset contains 398 instances with 9 attributes. Data collection for the original dataset was conducted through automotive specifications and performance testing during the 1970-1982 period.

3.3. Structure of Variables

The dataset variables are categorized as follows:

Numerical Variables (Continuous):

- Miles per gallon (mpg)
- Engine displacement (cubic inches)
- Horsepower
- Weight (pounds)
- Acceleration (seconds)

Categorical Variables (Discrete):

- Number of cylinders (4, 6, 8)
- Model year (1970-1982)
- Origin (USA, Europe, Japan)
- Car brand

For statistical analysis, categorical variables will be examined using frequency distributions, contingency tables, and chi-squared tests. Numerical variables will be analyzed using measures of central tendency, dispersion, Parametric tests, and hypothesis testing.

3.4 Preliminary Data Analysis

3.4.1 Descriptive Analysis

Descriptive analysis under preliminary data analysis is carried out to identify characteristics of data, patterns of data distributions, and mainly to discover anomalies in data collected before going deeper into analyzing all the data and making conclusions. Categorical and numerical data analysis techniques will be used in descriptive analysis. As categorical techniques contingency tables, frequency tables and pie charts will be used while as numerical techniques mean, median, mode, range, variance, standard deviation and bar graphs will be used.

3.4.2 Demographic Data Analysis

Demographic analysis is conducted to determine how the demographic data are distributed throughout the sample used in this study. Demographic data can be described as attributes that define the population of vehicles in the Auto-MPG dataset. It describes characteristics within this population. For this analysis, model year, country of origin, car brand and number of cylinders are considered as demographic attributes.

Distribution of Origin

The origin attribute categorizes vehicles by manufacturing region: USA (1), Europe (2), and Japan (3)

Pie Chart of Car Origins (Sample of 50)

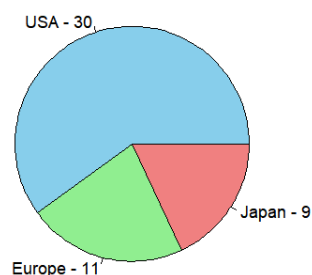


Figure 3.4.2.1: Pie Chart of Origin

Origin	Frequency	Percentage
USA	30	60%
Europe	11	22%
Japan	9	18%

Table 3.4.2.1: Origin Table

The sample maintains reasonable representation across origins, with American vehicles comprising more than half the sample, reflecting the dataset's composition during a period when domestic manufacturers dominated the US market.

Distribution of Model Year

The *model year* attribute represents the manufacturing year of each vehicle, ranging from 1970 to 1982.

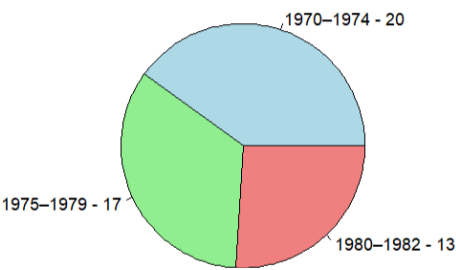


Figure 3.4.2.2: Pie Chart Model Year

Year Range	Frequency	Percentage
1970-1974	20	40%
1975-1979	17	34%
1980-1982	13	26%

Table 3.4.2.2: Model Year Table

The sample provides relatively even distribution across the 13-year period, enabling temporal trend analysis of fuel efficiency improvements.

Distribution of Cylinders

Engine cylinder count represents engine size and power capacity.

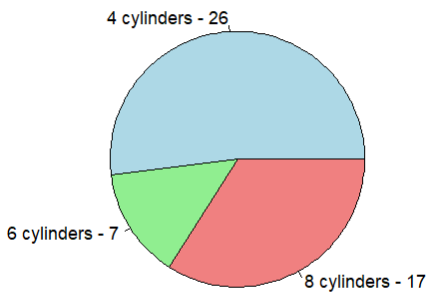


Figure 3.4.2.3: Pie Chart Cylinders

Cylinders	Frequency	Percentage
4	26	52%
6	7	14%
8	17	34%

Table 3.4.2.3: Cylinders Table

The distribution shows a shift toward smaller 4-cylinder engines, consistent with the fuel economy priorities emerging during the 1970s oil crises.

Distribution of Car Brands

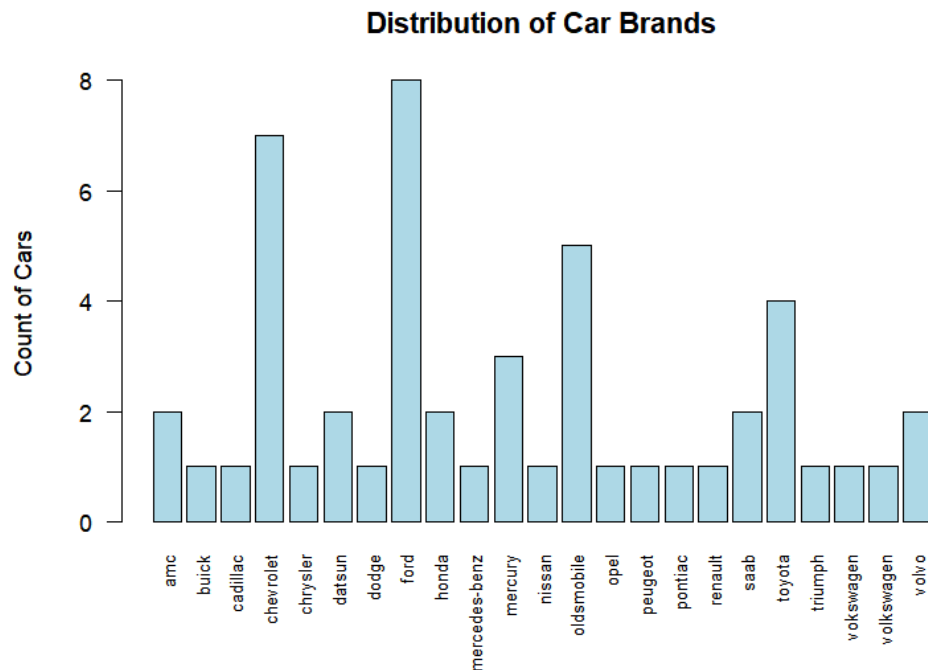


Figure 3.4.2.4 Distribution of Car Brands

The distribution shows that certain brands like Ford, Chevrolet, and Datsun appear more frequently in the sample, reflecting their popularity and production dominance during the 1970s. The variety of brands also indicates a mix of American and international manufacturers, aligning with the global expansion of the automobile market during that era.

3.5 Descriptive data analysis

Under this chapter all the categorical and numerical attributes used in this survey will be described, analyzed and summarized. Furthermore, inferential statistic methods using the data collected will be presented in this chapter.

Attributes used in the survey.

Categorical	Numerical
• Cylinders • Model Year • Origin • Car brand	• mpg - Miles per gallon (fuel efficiency) • displacement - Engine displacement in cubic inches • horsepower - Engine horsepower • weight - Vehicle weight in pounds • acceleration - Time to accelerate 0-60 mph in seconds

Table :3.5.1 Data categories

Categorical Attributes

Cylinders, Model Year, Origin, Car brands will not be discussed under this chapter since they were already described in chapter 3.4.2. Demographic data analysis.

Numerical Attributes

Measures the distance a vehicle can travel per unit of fuel, serving as the primary indicator of fuel economy.

Methods Used in Describing Numerical Data

1. **Sample Arithmetic Mean** - Average numerical value of data distribution (expected values). Summation of all values divided by the sample size.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Figure 3.5.1: Mean

2. **Mode** - The most frequent numerical value in data distribution.

3. **Median** - Middle value of an ordered numerical data distribution where 50% of values are greater than this and 50% of values are less than this (array starts with 1). When the sample size is odd, the median is simply considered as the value in the middle position. However, it becomes slightly different when it comes to an even size sample size.

i. **Finding the position (sample size is odd)**

$$\frac{n + 1}{2}$$

Figure 3.5.2: Median (position)

ii. **Median (Sample size is even)**

i. Position should be obtained. (x = position value)

ii. Mode = x position value + (x+0.5) position value - x position value)

4. **Range** - It's the difference between the highest value of the distribution and the lowest value.

Range = Highest value - Lowest value

5. **Variance** - It's the measurement of the spread between numbers in a data set. It represents how the values are spread from the mean.

$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Figure 3.5.3: Variance

6. **Standard Deviation** - It's the measure of how dispersed the data is in relation to the mean.

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Figure 3.5.4: Standard Deviation

Miles Per Gallon (MPG) Distribution

This is a measure of the vehicle's fuel efficiency. It represents the distance (in miles) a car can travel using one gallon of fuel. Higher mpg values indicate better fuel economy.

10.0, 12.0, 12.0, 13.0, 13.0, 14.0, 14.0, 14.0, 14.0, 15.0, 15.0, 15.5, 16.0, 16.5, 16.5, 17.5, 18.0, 18.1, 19.2, 19.4, 21.0, 21.0, 23.9, 24.0, 24.5, 25.0, 25.0, 25.0, 25.1, 26.0, 26.0, 26.6, 26.8, 28.0, 28.0, 29.0, 29.8, 29.9, 30.7, 31.0, 33.0, 35.0, 36.0, 36.0, 36.0, 36.1, 38.0, 38.0, 38.1, 44.6

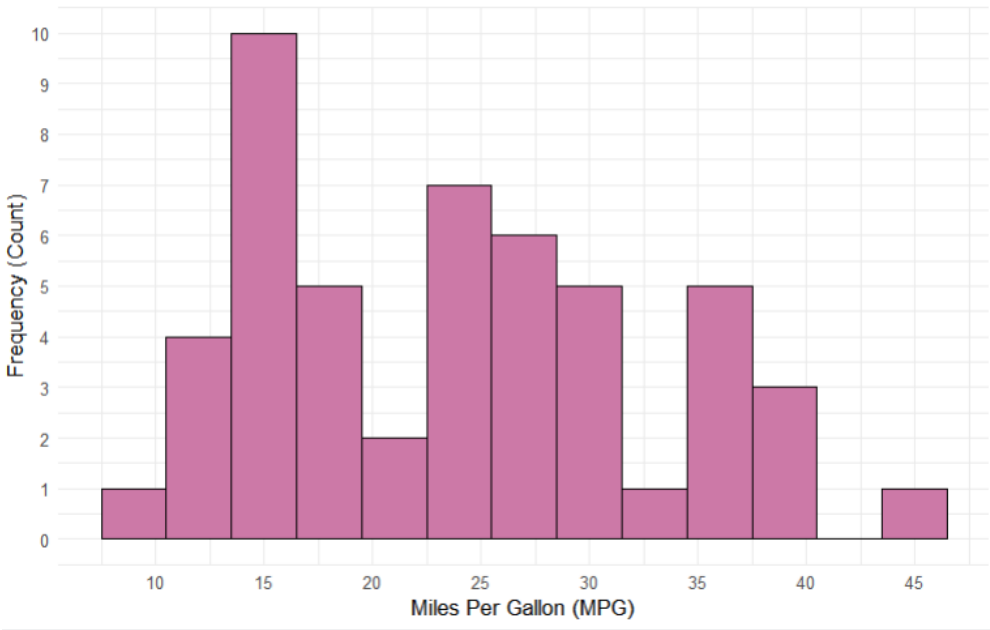


Figure 3.5.5: Bar graph of MPG

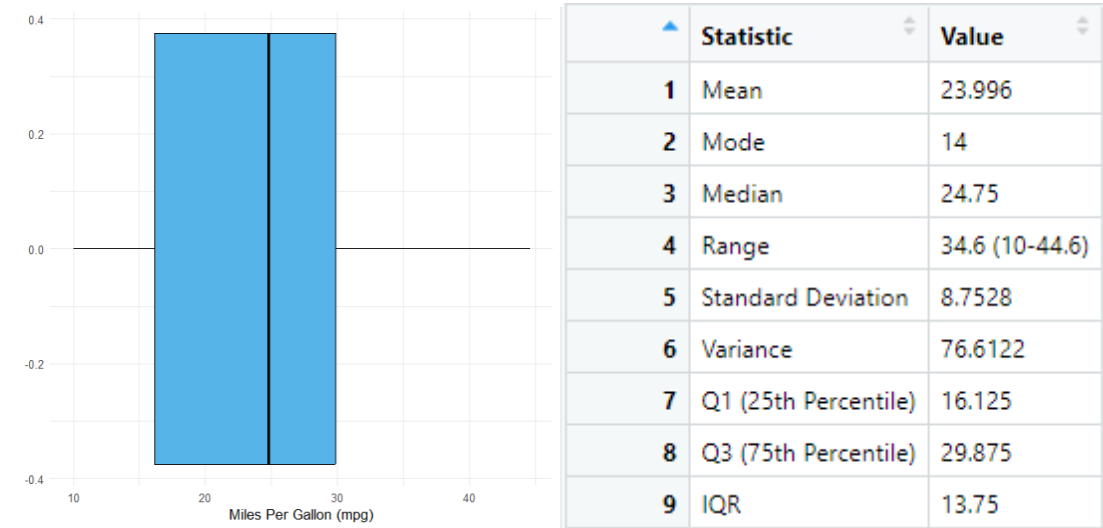


Figure 3.5.6: Box Plot of MPG

Table 3.5.2: MPG table

Displacement (Engine displacement in cubic inches) Distribution

The total volume is swept by all the pistons in the cylinders of an engine. It is a key factor in determining an engine's potential power output. Larger displacement generally correlates with more powerful engines and lower.

76, 79, 79, 89, 89, 91, 91, 91, 97, 97, 98, 98, 98, 98, 104, 113, 116, 120, 120, 120, 121, 121, 122, 140, 140, 145, 156, 168, 173, 232, 250, 250, 260, 262, 267, 302, 302, 304, 318, 350, 350, 350, 350, 360, 400, 400, 400, 429, 454, 455

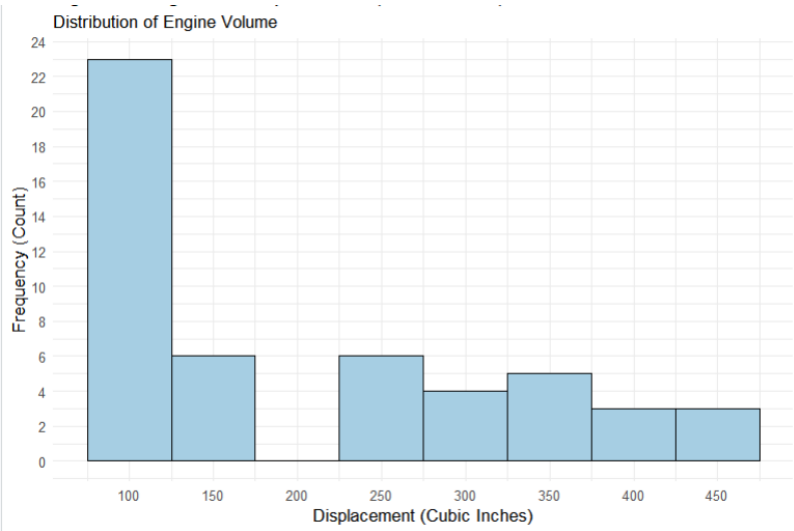


Figure 3.5.7: Bar graph of Displacement

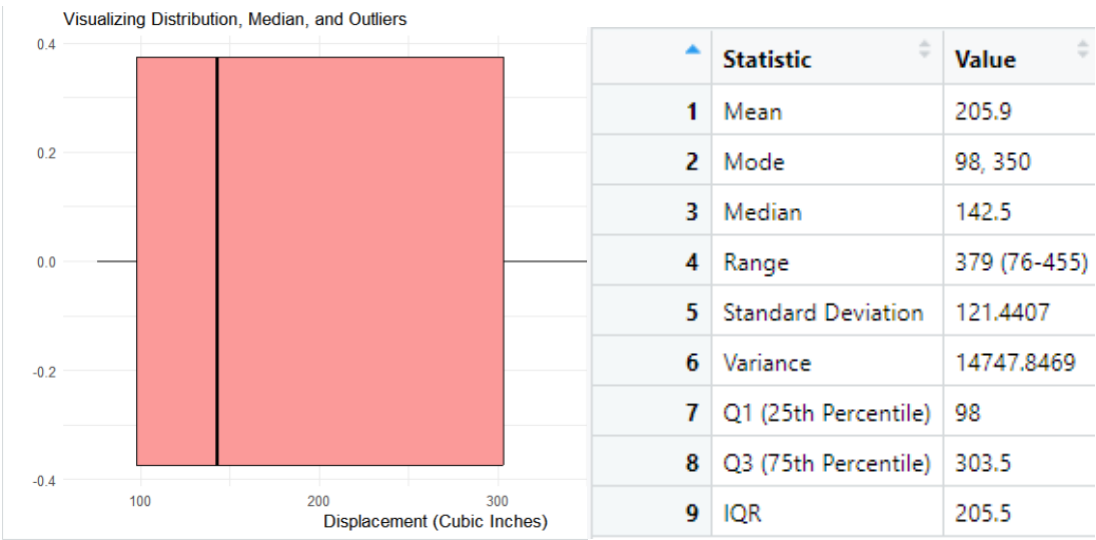


Figure 3.5.8: Box Plot of Displacement

Table 3.5.3: Displacement table

Horsepower (Engine horsepower)

A measure of the engine's power output—the rate at which the engine can do work. It is directly related to the vehicle's performance capabilities, such as acceleration and top speed.

52, 53, 58, 60, 60, 62, 65, 66, 67, 67, 67, 70, 72, 72, 75, 75, 76, 79, 85, 87, 88, 88, 88, 90, 90, 92, 92, 95, 95, 105, 110, 112, 115, 115, 120, 125, 139, 140, 150, 150, 150, 160, 170, 170, 175, 180, 198, 215, 220, 225

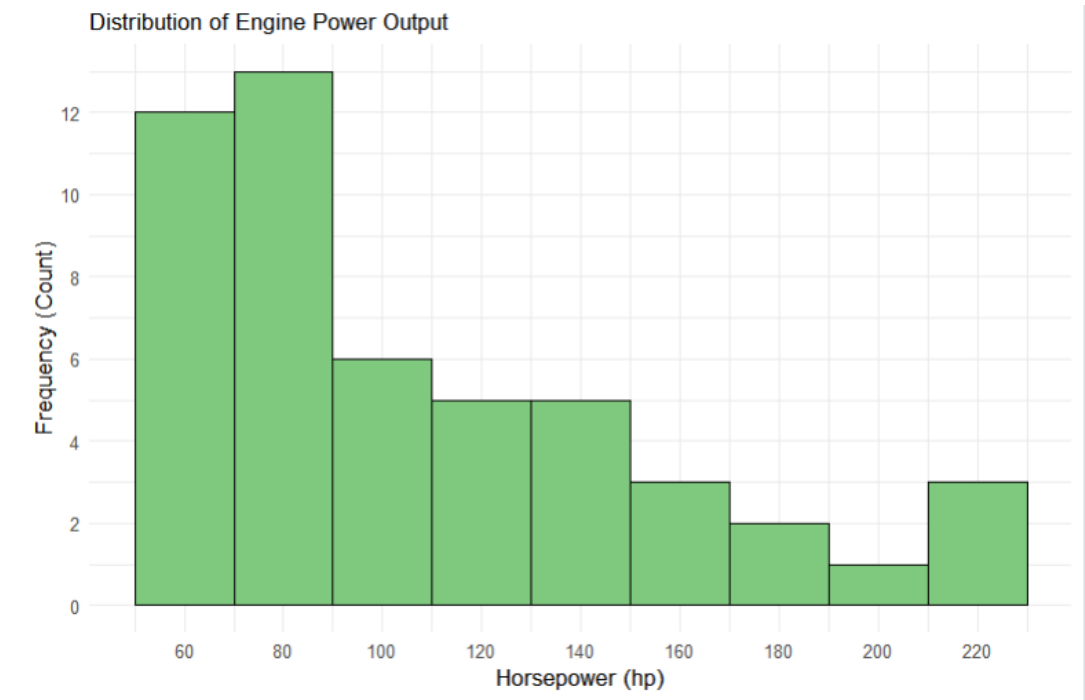


Figure 3.5.9: Bar graph of horsepower

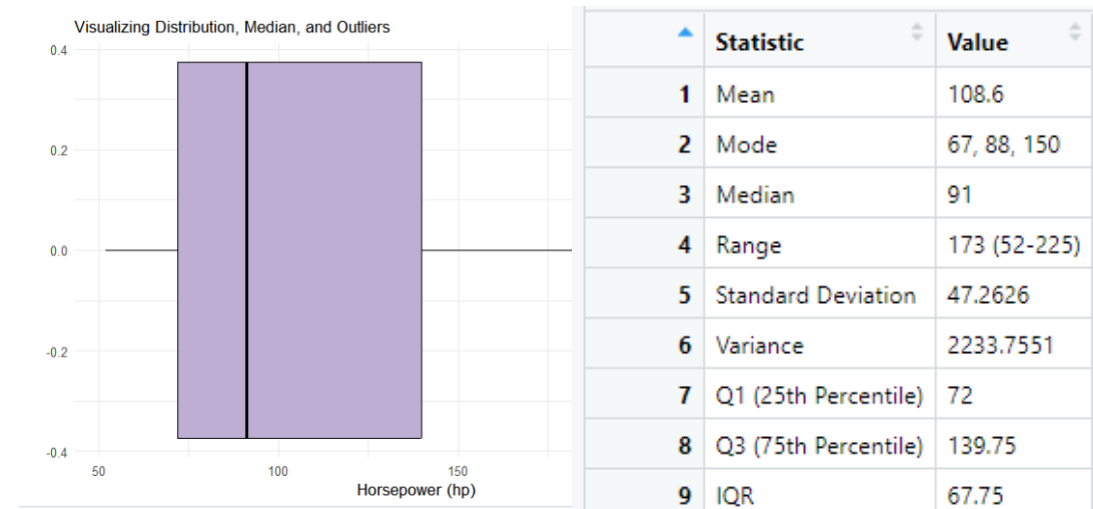


Figure 3.5.10: Box Plot of horsepower

Table 3.5.4: Horsepower table

Weight (Vehicle weight in pounds)

The curb weight of the car. Heavier vehicles require more energy to move, leading to a strong inverse relationship with mpg (i.e., higher weight means lower fuel efficiency).

1649, 1795, 1800, 1825, 1845, 1850, 1963, 1968, 1995, 2125, 2158, 2160, 2164, 2171, 2228, 2288, 2375, 2380, 2401, 2500, 2585, 2625, 2671, 2700, 2720, 2933, 2979, 3015, 3086, 3160, 3205, 3210, 3420, 3432, 3520, 3605, 3725, 3820, 3892, 4141, 4165, 4354, 4380, 4385, 4456, 4457, 4464, 4615, 4746, 4952

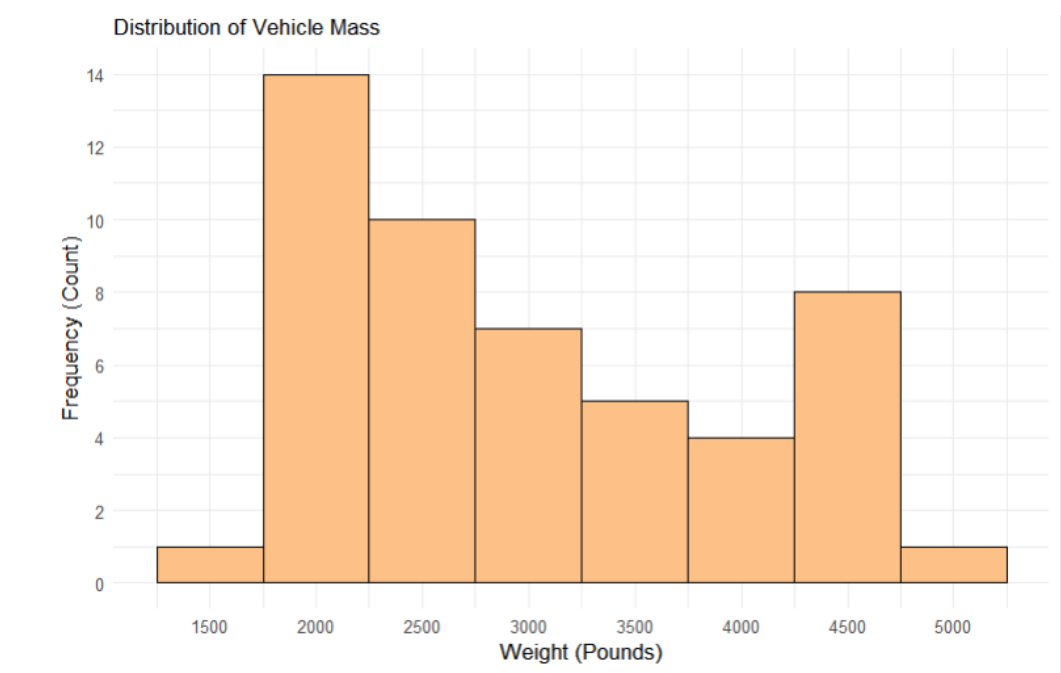


Figure 3.5.11: Bar graph of Weight

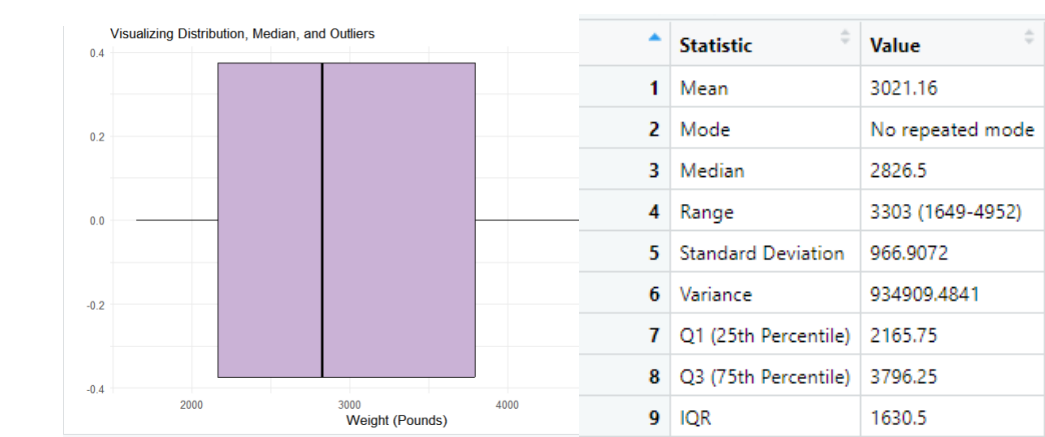


Figure 3.5.12: Box Plot of Weight

Table 3.5.5: Weight table

Acceleration (Time to accelerate)

The time (usually in seconds) it takes for the car to accelerate from a standstill to a specific speed (often 60 mph or 100 km/h). A lower value indicates faster acceleration and, typically, a more powerful and performance-oriented vehicle.

9.0, 10.0, 11.2, 11.4, 11.5, 12.0, 12.0, 12.0, 12.1, 12.5, 12.9, 13.5, 13.5, 13.5, 13.8, 14.0, 14.0, 14.0, 14.4, 14.5, 14.5, 14.5, 15.0, 15.1, 15.3, 15.4, 15.5, 15.5, 16.0, 16.2, 16.4, 16.5, 16.7, 17.0, 17.0, 17.2, 17.3, 17.4, 17.5, 18.6, 18.6, 18.8, 19.0, 19.5, 19.5, 19.6, 20.7, 21.0, 22.1, 22.2

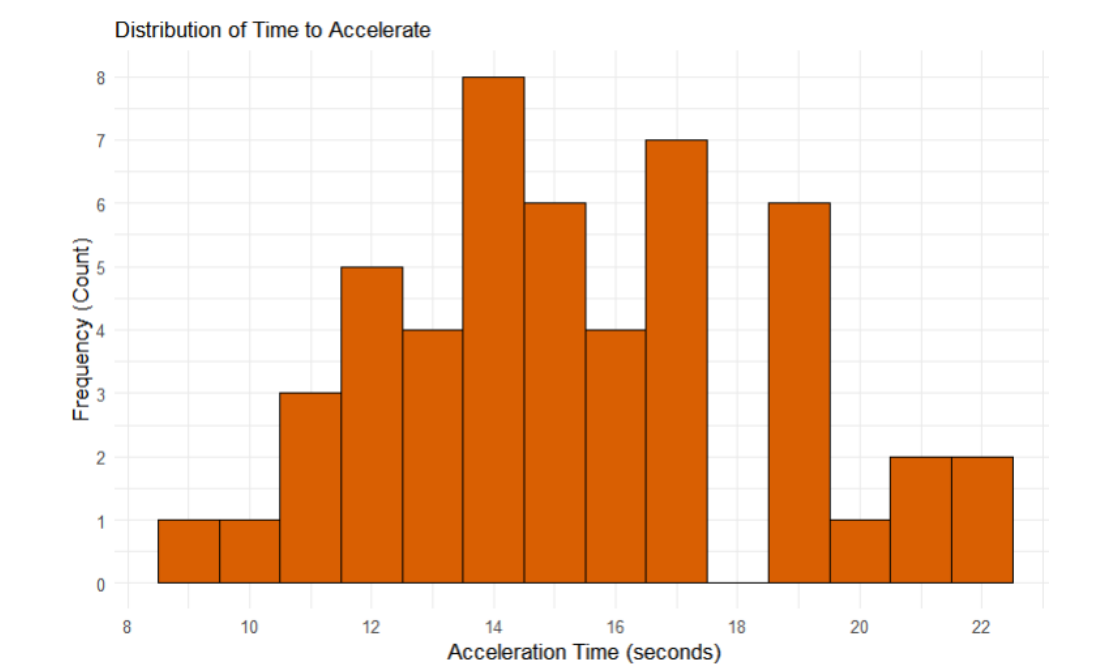


Figure 3.5.13: Bar graph of Acceleration

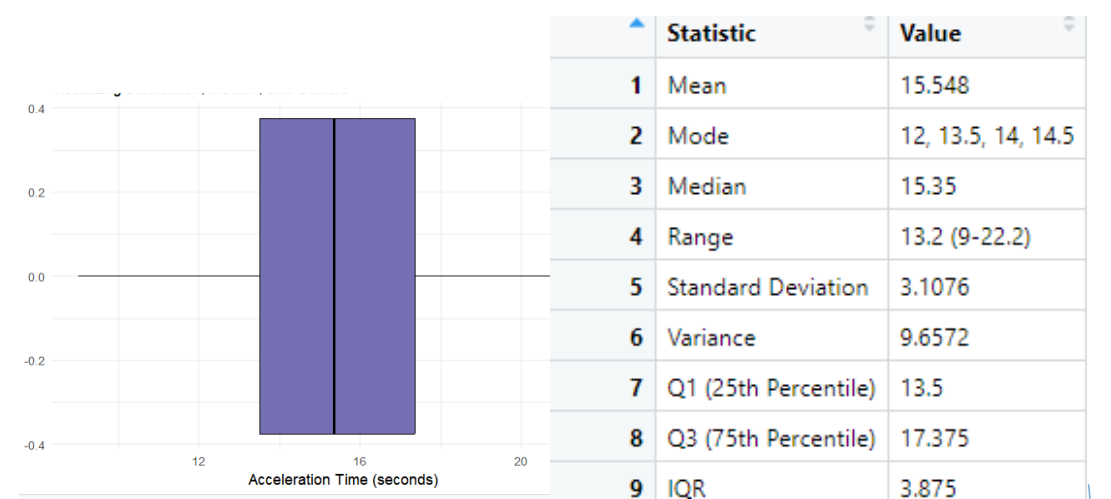


Figure 3.5.14: Box Plot of Acceleration

Table 3.5.6: Acceleration table

Chapter 4-Results

4.1 Inferential Data Analysis

This chapter presents the inferential statistical analysis conducted to examine the factors influencing automobile fuel efficiency using a sample of 50 vehicles from the Auto-MPG dataset. The analysis focuses on determining whether fuel efficiency (measured in miles per gallon – MPG) differs significantly across key vehicle characteristics such as:

- Cylinder count
- Vehicle weight
- Horsepower
- Engine displacement
- Region of origin

Both **Welch's two-sample t-tests** and a **Chi-square test of independence** were used to test hypotheses.

4.1.1 Assumptions for Inferential Tests

Before conducting inferential tests, the following assumptions were applied:

t-Test Assumptions

To ensure statistical validity, the following assumptions were considered:

1. **Normality:**
With sample sizes ≥ 25 per group, the sampling distribution of the mean is approximately normal by the Central Limit Theorem.
2. **Independence:**
Each vehicle represents an independent observation.
3. **Variance Assumption:**
Equal variances are **not** assumed. Therefore, **Welch's t-test** was used for all mean comparisons.

Chi-Square Test Assumptions

- Categorical variables are mutually exclusive.
- Observations are independent.

These assumptions follow standard statistical practices for automotive data analysis.

4.2 Hypothesis Testing Using Welch's t-test

The general form of the Welch t-statistics is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Figure 4.1 Welch t-statistic Formula

- $\bar{x}_1, \bar{x}_2$: sample means
- s_1^2, s_2^2 : sample variances
- n_1, n_2 : sample sizes

Degrees of freedom:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

Figure 4.2 Degrees of Freedom Formula

Level of significance = 0.05 (standard level for inferential statistics in automotive studies).

Reject H_0 if P-value < level of significance.

1970-1982 automobiles are considered as the population parameter in this scenario.

- Below the average MPG for 8-cylinder vehicles will be considered low efficiency, and above the average for 4-cylinder vehicles will be considered high efficiency.

Q1. Do vehicles with fewer cylinders have higher fuel efficiency?

Null Hypothesis – There is no significant difference in mean MPG between 4-cylinder and 8-cylinder vehicles.

H_0 : mean MPG (4-cyl) = mean MPG (8-cyl)

Alternative Hypothesis – There is a significant difference in mean MPG between 4-cylinder and 8-cylinder vehicles (4-cylinder vehicles are more efficient).

H_1 : mean MPG (4-cyl) > mean MPG (8-cyl)

```

> orderCyl4 <- sort(subset(auto_sample$mpg, auto_sample$cylinders == 4))
> orderCyl8 <- sort(subset(auto_sample$mpg, auto_sample$cylinders == 8))
> resultCyl <- t.test(orderCyl4, orderCyl8, mu=0, conf.level=0.95, alternative = "greater", var.equal = FALSE)
> resultCyl

Welch Two Sample t-test

data: orderCyl4 and orderCyl8
t = 8.5196, df = 40.971, p-value = 6.518e-11
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 11.14357      Inf
sample estimates:
mean of x mean of y
 29.58077  15.69412

```

Figure 4.3: T-Test Output: Effect of Cylinder Count on MPG

The data obtained from the t-test indicates:

- **P-Value of Null hypothesis being true** = $6.18 \times 10^{-11} = 0.0000000000618$

Since the level of significance is 0.05, p-value obtained from the test is less than the level of significance.

p-value < Level of significance = $6.18 \times 10^{-11} < 0.05$

Hence H_0 can be rejected.

Cylinder Count	Count	Mean MPG	Standard Deviation
4	26	29.58	6.42
8	17	15.69	4.27

Table 4.1: Effect of Cylinder Count on Fuel Efficiency

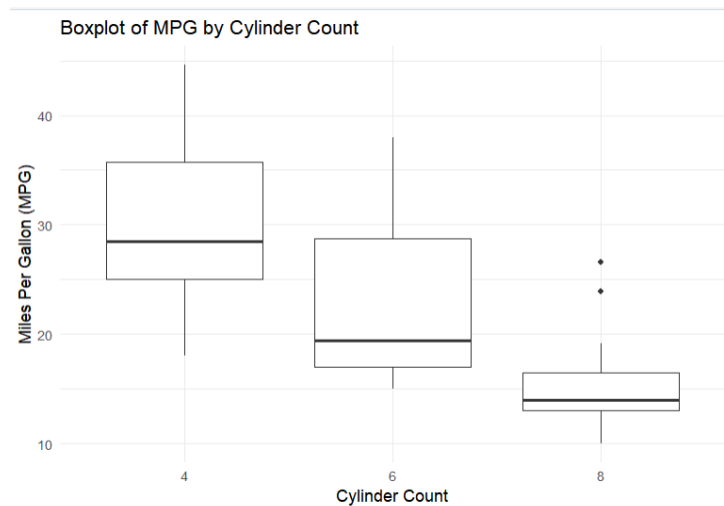


Figure 4.4: Box Plot: Effect of Cylinder Count on MPG

Conclusion – Vehicles with 4 cylinders have significantly higher mean MPG (29.58 mpg) than those with 8 cylinders (15.69 mpg).

The mean MPG of 4-cylinder vehicles is significantly higher than that of 8-cylinder vehicles. This aligns with the expectation that smaller engines are more fuel efficient. The difference is both statistically and practically significant.

Q2. Is there a significant relationship between vehicle weight and fuel efficiency?

Null Hypothesis (H₀): There is no significant difference in mean mpg between vehicles with weight below the median and those above the median.

- H₀: mean mpg (low weight) = mean mpg (high weight)

Alternative Hypothesis (H₁): Vehicles with weight below the median have significantly higher mean mpg than those with weight above the median.

- H₁: mean mpg (low weight) > mean mpg (high weight)

```
> print(result_weight)

Welch Two Sample t-test

data: low_weight and high_weight
t = 7.2879, df = 47.685, p-value = 1.382e-09
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 9.669084      Inf
sample estimates:
mean of x mean of y
 30.276    17.716
```

Figure 4.5: T-Test Output: Effect of Weight on MPG

- The data obtained from the Welch two-sample t-test indicates:
- Low Weight Group: $n \approx 25$, mean mpg = 30.28, sd ≈ 6.50
- High Weight Group: $n \approx 25$, mean mpg = 17.72, sd ≈ 5.00
- t-test: $t = 7.29$, $df = 47.69$, $p\text{-value} = 1.382 \times 10^{-9}$

weight group	Count	Mean MPG	Standard Deviation
Low Weight	25	30.28	6.50
High Weight	25	17.72	5.00

Table 4.2: Effect of weight and fuel efficiency

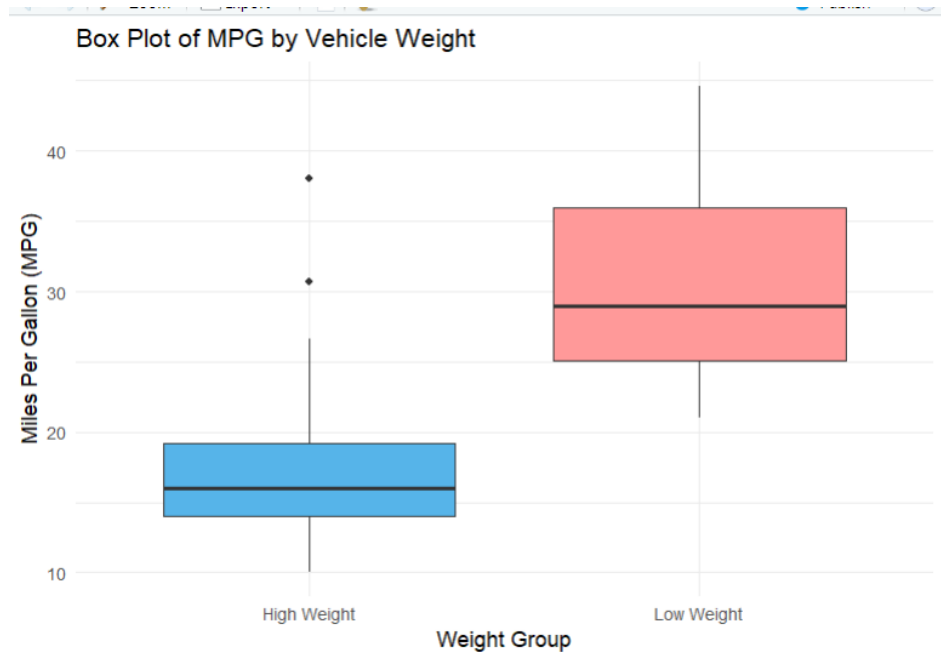


Figure 4.6: Boxplot of MPG Vs Weight

The data obtained from the t-test indicates:

The results of the Welch's t-test show a **p-value of 1.382×10^{-9}** , which is far below the significance level of **$\alpha = 0.05$** . Since

$$p\text{-value} = 1.382 \times 10^{-9} < 0.05,$$

The null hypothesis is rejected.

Conclusion:

Vehicles with weight below the median demonstrate **significantly higher fuel efficiency** (mean = 30.28 mpg) compared to vehicles above the median (mean = 17.72 mpg). This confirms that lighter vehicles achieve better fuel economy, consistent with engineering principles that reduced mass requires less energy for acceleration. Weight is therefore one of the strongest predictors of fuel efficiency within this dataset.

Q3. Is there a significant relationship between horsepower and mpg?

Null Hypothesis (H_0): There is no significant difference in mean mpg between vehicles with horsepower below the median and those above the median.

- H_0 : mean mpg (low horsepower) = mean mpg (high horsepower)

Alternative Hypothesis (H_1): Vehicles with horsepower below the median have significantly higher mean mpg than those with horsepower above the median.

- H_1 : mean mpg (low horsepower) > mean mpg (high horsepower)

```
> print(result_hp)

Welch Two Sample t-test

data: low_hp and high_hp
t = 6.582, df = 45.131, p-value = 2.102e-08
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 8.890638      Inf
sample estimates:
mean of x mean of y
 29.964    18.028
```

Figure 4.7 T-Test Output: Effect of Horsepower on MPG

t-test result:

- $t = 6.582$, $df = 45.131$, $p\text{-value} = 2.102e-08$
- Since $p < 0.05$, reject H_0 .

Horsepower Group	Count	Mean MPG	Standard Deviation
Low Horsepower	25	29.96	7.17
High Horsepower	25	18.3	5.54

Table 4.3: Effect of Horsepower on Fuel Efficiency

```
<
> # Summary statistics
> hp_stats <- auto_sample %>%
+   mutate(hp_group = ifelse(horsepower <= median_hp, "Low Horsepower", "High Horsepower")) %>%
+   group_by(hp_group) %>%
+   summarise(n = n(), mean_mpg = round(mean(mpg, na.rm = TRUE), 2),
+             sd_mpg = round(sd(mpg, na.rm = TRUE), 2))
> print(hp_stats)
# A tibble: 2 x 4
  hp_group      n mean_mpg sd_mpg
  <chr>    <int>    <dbl> <dbl>
1 High Horsepower 25    18.0   5.54
2 Low Horsepower  25    30.0   7.17
> |
```

Figure 4.8: Hypothesis Testing: Effect Of Horsepower on Mpg

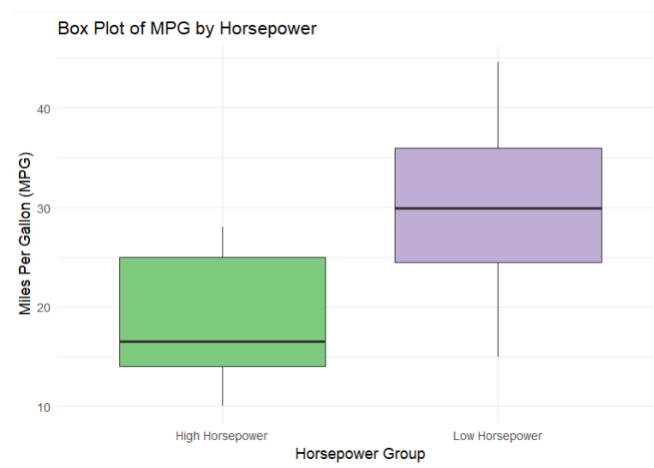


Figure 4.9: Boxplot of MPG vs Horsepower

The data obtained from the t-test indicates:

The Welch's t-test produced a p-value of 2.102×10^{-8} , which is far below the significance level of $\alpha = 0.05$. Since

$$p\text{-value} = 2.102 \times 10^{-8} < 0.05,$$

The null hypothesis is rejected.

Conclusion:

Vehicles with horsepower below the median exhibit significantly higher fuel efficiency (mean = 30.00 mpg) compared to vehicles with above-median horsepower (mean = 18.03 mpg). This confirms that horsepower has a strong influence on fuel consumption higher horsepower engines require more fuel to generate greater power output, resulting in lower MPG. Thus, lower horsepower vehicles are substantially more fuel-efficient.

Q4. Is there a significant relationship between engine displacement and mpg?

Null Hypothesis (H_0): There is no significant difference in mean mpg between vehicles with displacement below the median and those above the median.

- H_0 : mean mpg (low displacement) = mean mpg (high displacement)

Alternative Hypothesis (H_1): Vehicles with displacement below the median have significantly higher mean mpg than those above the median.

- H_1 : mean mpg (low displacement) > mean mpg (high displacement)

```
> print(result_disp)

Welch Two Sample t-test

data: low_disp and high_disp
t = 6.1039, df = 47.933, p-value = 8.74e-08
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 8.308045      Inf
sample estimates:
mean of x mean of y
 29.724    18.268
```

Figure 4.10 T-Test Output: Effect of Engine Displacement on MPG

t-test result:

- $t = 6.104$, $df = 47.933$, $p\text{-value} = 8.74e-08$
- Since $p < 0.05$, reject H_0 .

Displacement Group	Count	Mean MPG	Standard Deviation
Low Displacement	25	29.7	6.51

High Displacement	25	18.3	6.76
-------------------	----	------	------

Figure 4.4: Effect of Engine Displacement on Fuel Efficiency

```

>
> # Summary statistics
> disp_stats <- auto_sample %>%
+   mutate(displacement = ifelse(displacement <= median_disp, "Low Displacement", "High Displacement")) %>%
+   group_by(displacement) %>%
+   summarise(n = n(), mean_mpg = round(mean(mpg, na.rm = TRUE), 2),
+           sd_mpg = round(sd(mpg, na.rm = TRUE), 2))
> print(disp_stats)
# A tibble: 2 x 4
  displacement    n mean_mpg sd_mpg
  <chr>          <int>    <dbl> <dbl>
1 High Displacement    25     18.3   6.76
2 Low Displacement    25     29.7   6.51

```

Figure 4.11: T-Test Output: Effect Engine Displacement And Mpg

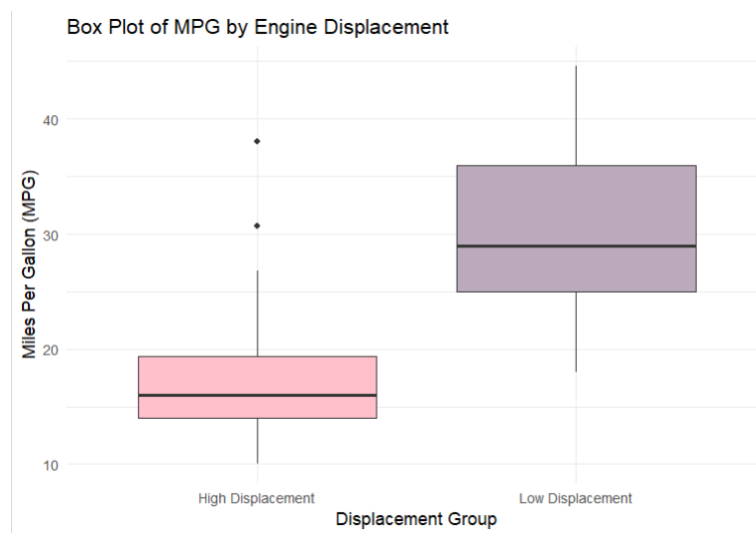


Figure 4.12: Boxplot Of MPG Vs Displacement

The data obtained from the t-test indicates:

The Welch's t-test resulted in a p-value of 8.74×10^{-8} , which is far below the significance threshold of $\alpha = 0.05$. Since

$$p\text{-value} = 8.74 \times 10^{-8} < 0.05,$$

The null hypothesis is rejected.

Conclusion:

Vehicles with engine displacement below the median demonstrate significantly higher fuel efficiency (mean = 29.72 mpg) compared to those with above-median displacement (mean = 18.27 mpg). This confirms that engine displacement has a substantial impact on fuel consumption. Lower displacement engines require less fuel to operate, resulting in consistently higher MPG values and indicating superior fuel efficiency.

Chi-Square Test

Chi-square distribution is used.

Level of significance = 0.05 (standard threshold for statistical significance).

Reject H_0 if p-value < level of significance.

Q5. Is there an association between the cylinder count and the origin of vehicles?

Null Hypothesis – There is no association between the cylinder count and the origin of vehicles.

H_0 : Cylinder count and origin are independent.

Alternative Hypothesis – There is an association between the cylinder count and the origin of vehicles.

H_1 : Cylinder count and origin are not independent.

```
> View(chi_result)
> chi_result

Pearson's Chi-squared test

data: contingency_table
X-squared = 22.454, df = 4, p-value = 0.0001627
.
```

Figure 4.13: Chi squared: Effect engine cylinder count and the origin

Observed Frequencies:

The following contingency table displays the observed counts of vehicles for each combination of cylinder count and origin.

Cylinders	USA	Europe	Japan	Total
4	10	7	9	26
6	5	2	0	7
8	15	2	0	17
Total	30	11	9	50

Figure 4.5: Observed Frequency Table

Expected Frequencies:

The expected frequencies, calculated under the assumption of independence, are presented in the table below. The formula used was:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{50}$$

Figure 4.14 Formula for expected frequency

Cylinders	USA	Europe	Japan	Total
4	15.60	5.72	4.68	26
6	4.20	1.54	1.26	7
8	10.20	3.74	3.06	17
Total	30	11	9	50

Figure 4.6: Expected Frequency Table

The data obtained from the chi-square test indicates:

The results of the Chi-Square test were as follows,

$$X^2(4, N = 50) = 22.454, p = .00016$$

Since the obtained p-value (.00016) is less than the level of significance ($\alpha = .05$), we reject the null hypothesis.

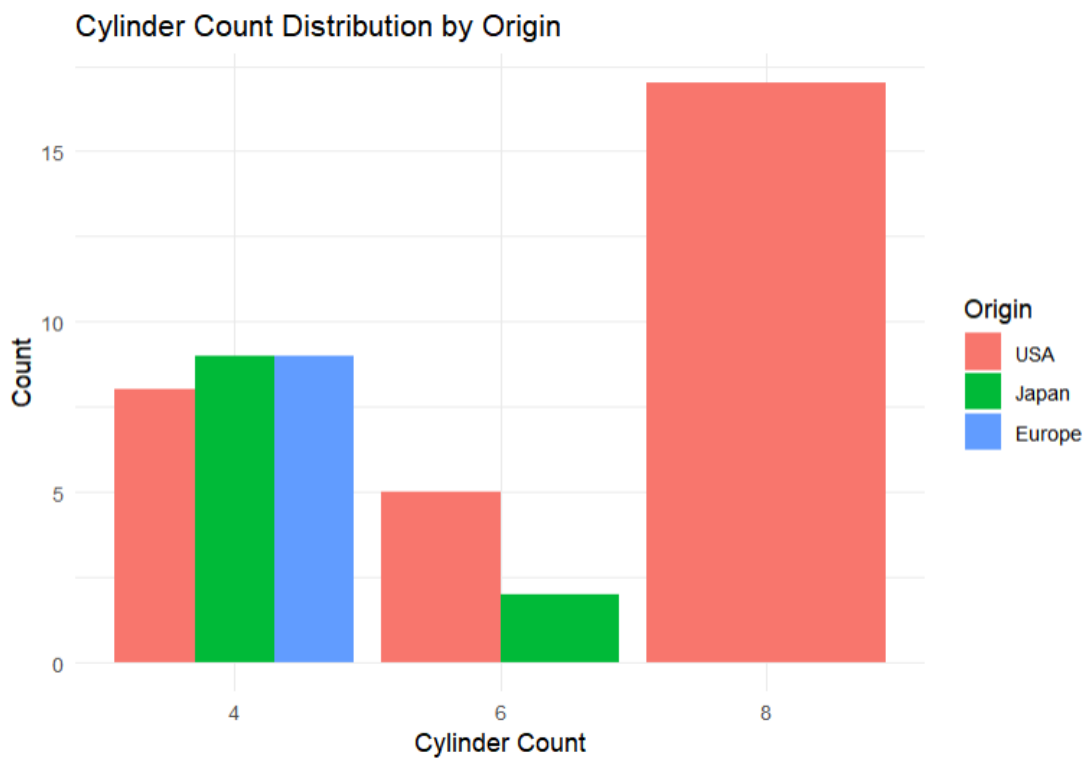


Figure 4.15: Bar graph of engine cylinder count and the origin

The null hypothesis is rejected.

Conclusion – There is statistically significant evidence to conclude that an association exists between engine cylinder count and the region of origin of vehicles. The distribution of cylinder counts is not independent of the manufacturing region.

The bar graph (Figure 4.15) and the observed frequency table illustrate this relationship:

- Japan exclusively produced 4-cylinder engines in this sample.
- The USA had a strong preference for larger 8-cylinder engines, while also manufacturing a substantial number of 4 and 6-cylinder vehicles.
- Europe showed a primary focus on 4-cylinder engines, with some 6 and 8-cylinder models.

This pattern reflects the distinct automotive manufacturing philosophies and market demands during the 1970-1982 period, heavily influenced by the global fuel crisis. Japanese and European manufacturers prioritized fuel efficiency with smaller engines, whereas American manufacturers maintained a stronger presence in the market for larger, more powerful vehicles.

Chapter 5 – Discussion and Conclusions

This study examined the Auto-MPG dataset, which comprises data on automobiles manufactured from 1970 to 1982—a transformative period influenced by global oil crises that fundamentally reshaped priorities in vehicle design and production. The extensive analysis of this dataset revealed that a variety of vehicle characteristics played pivotal roles in determining fuel efficiency during this critical era.

A key finding is the substantial influence of engine cylinder count on fuel economy. Vehicles equipped with 4 cylinders exhibited significantly higher average miles per gallon compared to their 8-cylinder counterparts. This trend reflects the automotive industry's strategic shift towards smaller engines aimed at improving fuel economy in response to the energy shortages experienced in the 1970s. This downsizing reduced fuel consumption while maintaining sufficient power for typical driving conditions.

Vehicle weight emerged as another crucial factor impacting fuel efficiency. Lighter vehicles consistently demonstrated superior fuel economy relative to heavier models. This outcome is consistent with fundamental physical principles, as reduced mass lowers the energy required to propel a vehicle. Additionally, the industry's adoption of materials and design techniques targeting weight reduction during the oil crises underscores the practical relevance of this finding.

Engine displacement and horsepower also significantly influenced fuel consumption patterns. Cars with smaller engine displacements and lower horsepower ratings achieved better fuel economy, emphasizing the trade-offs manufacturers faced between engine power and efficiency. Regulatory frameworks and market demands during this period incentivized optimizing this balance, which had clear effects on vehicle performance and fuel use.

Regional differences in vehicle design and consequent fuel efficiency were evident. Japanese manufacturers predominantly produced compact, efficient vehicles that outperformed many American and European models in terms of fuel economy. This divergence likely stemmed from varying regional consumer preferences, regulatory environments, and engineering philosophies. Japanese cars' efficiency-oriented designs set benchmarks that influenced global automotive trends.

Temporal trends within the dataset reveal improvements in fuel efficiency over the 12-year period. Vehicles from later model years demonstrated enhanced fuel economy, indicative of technological advancements, improved engineering practices, and the impact of regulatory initiatives such as the Corporate Average Fuel Economy (CAFE) standards implemented in the mid-1970s. These measures encouraged manufacturers to continuously innovate in pursuit of greater fuel efficiency.

In sum, this study provides a detailed understanding of how engine size, vehicle weight, power characteristics, geopolitical factors, and evolving standards collectively shaped fuel efficiency outcomes in a formative period of automotive history. These insights contribute to a foundational knowledge base that informs not only historical perspectives but also provides contextual lessons for future developments in sustainable automotive design.

5.2 Limitations

The 50-car sample, drawn from a 398-car dataset, restricts the applicability of these findings. The data, limited to 1970–1982, excludes modern technologies like hybrids. Unverified t-test assumptions, such as normality, and the small sample size may affect result reliability. Factors like driving conditions were not considered, potentially influencing mpg variations

.

5.3 Recommendations and Future Studies

Automakers should focus on designs with fewer cylinders, lower weight, reduced horsepower, and smaller displacements to boost fuel efficiency. Policymakers could refine CAFE standards based on these trends. Future research should analyze the full dataset, include recent vehicle models, and assess additional factors like driving habits using t-tests or other statistical methods to enhance understanding.

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Appendices

R studio commands (Repetitive commands are not presented)

```
# Read the Auto-MPG data with proper parameters
auto <- read.table("autompg.data",
                  header = FALSE,
                  sep = ",",
                  quote = "\"",
                  na.strings = "?",
                  stringsAsFactors = FALSE)

colnames(auto) <- c("mpg", "cylinders", "displacement", "horsepower", "weight",
                  "acceleration", "model_year", "origin", "car_name")

head(auto)
set.seed(123)
sample_index <- sample(1:nrow(auto), 50)
auto_sample <- auto[sample_index, ]
write.csv(auto_sample, "auto_sample.csv", row.names = FALSE)
```

Figure 5.1: Reading the dataset extracting the sample

```
# Group model years into ranges
year_ranges <- cut(auto_sample$model_year,
                  breaks = c(69, 74, 79, 82),
                  labels = c("1970-1974", "1975-1979", "1980-1982"),
                  include.lowest = TRUE)

# Count cars by year range
year_range_count <- table(year_ranges)
cat("\n--- Model Year Range Counts ---\n")
print(year_range_count)

# Pie chart for model year distribution
pie(year_range_count,
    labels = paste(names(year_range_count), "-", year_range_count),
    col = c("lightblue", "lightgreen", "lightcoral"),
    main = "Pie Chart of Model Year Ranges (Sample of 50)")
```

Figure 5.2: creating pie chart for model year

```
mpg <- auto_sample$mpg
sorted_mpg_array <- sort(mpg)
print(sorted_mpg_array)

mean_mpg <- mean(mpg)
median_mpg <- median(mpg)
mode_mpg <- as.numeric(names(sort(table(mpg), decreasing = TRUE))[1])
range_mpg <- range(mpg)
sd_mpg <- sd(mpg)
var_mpg <- var(mpg)
q1_mpg <- quantile(mpg, 0.25)
q3_mpg <- quantile(mpg, 0.75)

ggplot(auto_sample, aes(x = mpg)) +
  geom_histogram(binwidth = 3, fill = "skyblue", color = "black") +
  labs(title = "Histogram of MPG", x = "Miles per Gallon", y = "Frequency") +
  theme_minimal()

ggplot(auto_sample, aes(y = mpg)) +
  geom_boxplot(fill = "lightgreen", color = "black") +
  labs(title = "Boxplot of MPG", y = "MPG") +
  theme_minimal()
```

Figure 5.3: descriptive analysis of MPG

```

# ----- T-Test: Weight -----
median_weight <- median(auto_sample$weight, na.rm = TRUE)
low_weight <- auto_sample$mpg[auto_sample$weight <= median_weight]
high_weight <- auto_sample$mpg[auto_sample$weight > median_weight]

result_weight <- t.test(low_weight, high_weight, mu = 0, conf.level = 0.95,
                        alternative = "greater", var.equal = FALSE)
print(result_weight)

ggplot(auto_sample, aes(x = factor(ifelse(weight <= median_weight, "Low Weight", "High Weight")), y = mpg)) +
  geom_boxplot(fill = c("#56B4E9", "#FF9999")) +
  labs(title = "MPG by Vehicle Weight Group", x = "Weight Group", y = "MPG") +
  theme_minimal()
# ----- T-Test: Horsepower -----

```

Figure 5.4: T-Test: Weight

```

contingency_table <- table(auto_sample$cylinders, auto_sample$origin)
print(contingency_table)

chi_result <- chisq.test(contingency_table)
print(chi_result)

ggplot(auto_sample, aes(x = factor(cylinders), fill = factor(origin))) +
  geom_bar(position = "dodge") +
  labs(title = "Cylinder Count Distribution by Origin",
       x = "Cylinder Count", y = "Count", fill = "Origin") +
  theme_minimal()

```

Figure 5.5: Chi-Square Test

```

# Install and load required packages
install.packages("ggplot2")
install.packages("dplyr")
install.packages("stringr")

library(ggplot2)
library(dplyr)
library(stringr)

```

Figure 5.6: Required packages

```

> mpg_summary_df <- data.frame(
+   Statistic = c("Mean", "Mode", "Median", "Range", "Standard Deviation", "Variance", "Q1 (25th)", "Q3 (75th)"),
+   MPG = c(mean_mpg, mode_mpg, median_mpg, paste(range_mpg, collapse = " - "), sd_mpg, var_mpg, q1_mpg, q3_mpg),
+   Displacement = c(mean_disp, mode_disp, median_disp, paste(range_disp, collapse = " - "), sd_disp, var_disp, q1_disp, q3_disp),
+   Horsepower = c(mean_hp, mode_hp, median_hp, paste(range_hp, collapse = " - "), sd_hp, var_hp, q1_hp, q3_hp),
+   Weight = c(mean_weight, mode_weight, median_weight, paste(range_weight, collapse = " - "), sd_weight, var_weight, q1_weight, q3_weight),
+   Acceleration = c(mean_acc, mode_acc, median_acc, paste(range_acc, collapse = " - "), sd_acc, var_acc, q1_acc, q3_acc)
+ )
>
> print(mpg_summary_df)

```

	Statistic	MPG	Displacement	Horsepower	Weight	Acceleration
1	Mean	23.996	205.9	108.6	3021.16	15.548
2	Mode	14	98	67	1649	12
3	Median	24.75	142.5	91	2826.5	15.35
4	Range	10 - 44.6	76 - 455	52 - 225	1649 - 4952	9 - 22.2
5	Standard Deviation	8.75284117138136	121.440713678632	47.2626184424944	966.907174490723	3.10761014548584
6	Variance	76.6122285714286	14747.8469387755	2233.75510204082	934909.484081633	9.65724081632653
7	Q1 (25th)	16.125	98	72	2165.75	13.5
8	Q3 (75th)	29.875	303.5	139.75	3796.25	17.375

Figure 5.7: Summery Statics